CTF-SRNet: A Cross-Modal Text and Feature Modulation Network for Remote Sensing Image Super-Resolution

Lin Feng, Yining Wang, Haoran Gao, Xinying Wang, Zhixiong Huang, and Shenglan Liu

***Abstract*—The remote sensing image super-resolution (RSISR) has achieved significant progress in recent years. However, effectively recovering fine-grained structural details while maintaining high-level semantic consistency remains a challenging problem. Most existing RSISR methods rely solely on visual information from low-resolution (LR) inputs, ignoring auxiliary semantic cues that are crucial for reconstructing complex scenes in remote sensing imagery. To address these limitations, we propose a novel cross-modal text and feature modulation network, termed CTF-SRNet, for RSISR. Unlike conventional approaches, the CTF-SRNet explicitly incorporates textual semantic information to guide the super-resolution (SR) process. The proposed framework consists of a multiscale channel-modulated image encoder for robust feature extraction, an image–text feature interaction block (ITFIB) that injects text-guided semantic priors into visual features, and a feature modulation block (FMB) that adaptively refines cross-modal representations to enhance structural detail recovery. In addition, a multilevel loss supervision strategy is employed to progressively constrain intermediate outputs, encouraging consistent texture reconstruction across different scales. Extensive experiments on multiple remote sensing benchmarks, including UC Merced, RSSCN7, Aerial Image Dataset (AID), Dataset for Object Detection in Aerial Images (DOTA), and DIOR datasets, demonstrate that CTF-SRNet consistently outperforms the state-of-the-art RSISR methods in terms of both quantitative metrics and visual quality, while maintaining a lightweight model design. Furthermore, the downstream classification experiments validate that the proposed cross-modal SR framework effectively enhances discriminative structural and semantic information, highlighting its practical value for real-world remote sensing applications.**

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***Index Terms*—Image–text feature interaction, image super-resolution (SR), remote sensing, text-guided semantic priors.**

I. Introduction

THE remote sensing images (RSIs) are obtained through satellites, uncrewed aerial vehicles, or airborne platforms and are widely used in various Earth observation tasks, such as object detection [1], [2], [3], [4], urban planning [5], [6], disaster monitoring [7], and environmental surveillance [8]. However, due to limitations in sensor resolution, communication bandwidth, and atmospheric conditions, high-resolution (HR) RSIs are generally difficult to obtain, which somewhat restricts their applicability in high-precision fields.

The super-resolution (SR) techniques aim to reconstruct HR images from low-resolution (LR) inputs, thereby enhancing visual quality and detail. In recent years, the rapid development of convolutional neural networks (CNNs) [9] and transformers [10] has led to significant advancements in single image SR (SISR), achieving remarkable success in natural image SR tasks. However, directly applying these models to RSISR remains challenging. The RSIs typically exhibit more complex scene structures, larger scale variations, and richer texture patterns, which place higher demands on the model's ability to represent and reconstruct structural details. Additionally, the RSIs are often affected by acquisition noise, imaging distortions, and heterogeneous spatial distributions, which further complicate the SR reconstruction process.

However, the existing SR networks, despite their success in natural image tasks, still face significant limitations when applied to RSIs. The RSIs usually contain complex scene structures, diverse land-cover types, and rich texture variations, making accurate reconstruction more challenging. In addition, the conventional SR models mainly rely on LR visual observations and focus on pixel-level reconstruction, while often ignoring high-level semantic information. Due to severe information loss during degradation, semantically different scenes, such as parking lots, airport runways, and industrial roads, may exhibit similar LR patterns, leading to reconstruction ambiguity. As a result, the purely visual SR methods may generate blurred textures, incorrect structures, or semantically inconsistent details. Therefore, the cross-modal semantic priors are introduced to provide scene-level and object-level contextual guidance, helping the network recover more plausible structures and semantically consistent details for RSISR.

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To address the aforementioned challenges, this article proposes a novel RSISR framework, termed Cross-Modal Text and Feature Modulation Network for Remote Sensing Image Super-Resolution (CTF-SRNet). The proposed method aims to alleviate the limitations of the existing SR models in capturing complex structural details and maintaining semantic consistency in RSIs. Unlike conventional approaches that rely solely on visual information for reconstruction, CTF-SRNet introduces textual semantic priors into the SR process and employs cross-modal feature modulation to guide the reconstruction toward semantically consistent and structurally reliable results.

The proposed network consists of three major components: a multiscale channel-modulated image encoder, an image–text feature interaction module, and a progressive optimization-based image reconstruction module. In the image encoding stage, a multiscale channel modulation module is employed to extract discriminative visual representations from LR RSIs. This module adopts multiple depthwise convolution branches with different receptive fields to capture structures and textures at various spatial scales, while a channelwise weight learning mechanism is introduced to adaptively fuse multiscale features, enhancing the representation capability for complex remote sensing scenes.

To incorporate semantic guidance, textual descriptions are encoded into semantic feature vectors and further processed by a prompt predictor to generate adaptive textual prompts. These prompts are then aligned with visual features in the Contrastive Language-Image Pre-training (CLIP) semantic space, enabling effective cross-modal semantic guidance. Based on this alignment, an image–text feature interaction block (ITFIB) is constructed to facilitate deep interaction between image and text features. Within ITFIB, a feature modulation block (FMB) is designed to inject textual semantic information into image features through text-driven modulation. To mitigate potential semantic drift caused by direct modulation, a cross-attention-based fusion strategy is further employed to jointly model the original image features and the modulated features, ensuring that semantic guidance remains consistent with the underlying visual evidence.

In the reconstruction stage, a progressive optimization module (POM) is adopted to iteratively refine the super-resolved features. At each stage, the image–text interaction features are reintroduced, and intermediate SR outputs are generated under multilevel supervision. By jointly applying reconstruction, perceptual, and adversarial losses at different stages, the network progressively enhances structural fidelity and texture details, leading to HR RSIs.

In summary, the main contributions of this work are as follows.

1) We propose a CTF-SRNet for RSISR, which introduces textual semantic priors into the SR process and enables semantic-guided reconstruction within a unified framework.
2) We design a multiscale channel-modulated image encoder that captures structural and textural information at different spatial scales and adaptively fuses multiscale features, providing robust visual representations for complex remote sensing scenes.
3) We introduce an ITFIB equipped with an FMB, which injects textual semantic guidance into image features while employing cross-attention fusion to effectively alleviate semantic drift and preserve structural consistency.
4) We develop a progressive optimization strategy with multilevel supervision, which iteratively refines SR results and jointly optimizes reconstruction quality, perceptual fidelity, and texture realism through hierarchical loss constraints. The effectiveness and robustness of this strategy are extensively validated across multiple remote sensing benchmark datasets.

The remainder of this article is organized as follows. Section I reviews the related work on image SR, with a focus on remote sensing applications. Section III presents the detailed architecture and key components of the proposed CTF-SRNet. Section IV reports the experimental settings, datasets, evaluation metrics, and comparative results. Finally, Section V concludes this article and discusses future research directions.

II. Related Work

A. Natural Image SR

With the rapid advancement of deep learning, the natural image SR has achieved remarkable progress. Early deep learning-based SR methods primarily relied on CNNs. The SRCNN [11] was the first to formulate SR as an end-to-end regression problem using CNNs, demonstrating the feasibility of learning nonlinear mappings from LR to HR images. To improve computational efficiency and reconstruction accuracy, the FSRCNN [12] introduced a deeper architecture with deconvolution layers, enabling faster inference while maintaining high performance.

The subsequent works focused on increasing network depth and representation capacity. The VDSR [13], DRCN [14], and EDSR [15] adopted residual learning strategies, which effectively alleviated the vanishing gradient problem and facilitated the training of very deep networks. These residual-based models significantly improved SR performance by capturing more complex LR–HR mappings. Building upon this paradigm, the RCAN [16] proposed a residual-in-residual architecture combined with channel attention mechanisms, allowing the network to adaptively emphasize informative feature channels. The RFANet [17] further enhanced this idea by aggregating multilevel residual features, leading to more refined texture reconstruction.

More recently, the transformer-based architectures have emerged as a powerful alternative to CNNs due to their strong capability in modeling long-range dependencies. The SwinIR [18] introduced the Swin Transformer into SR by employing window-based self-attention, achieving the state-of-the-art results on multiple benchmarks. The HAT [19] further proposed a hybrid attention framework that combines channel attention, window-based self-attention, and overlapping cross-attention, effectively enhancing pixelwise interactions and improving reconstruction quality across diverse scenes. The omni-SR [20] introduces an omni-dimensional attention mechanism together with a multiscale feature aggregation strategy, allowing lightweight networks to deliver competitive and even superior SR performance. The OHT [21] introduces an occlusion-aware hybrid transformer that integrates occlusion maps with spatial–angular separable convolution and angular self-attention to enhance light field SR.

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B. SR for RSIs

Motivated by the success of deep learning-based SR methods for natural images, increasing research efforts have been devoted to design SR networks tailored to RSIs. Owing to the inherent properties of RSIs, such as complex land-cover textures, diverse imaging conditions, and large-scale spatial variations, the RSISR faces challenges that differ substantially from natural image SR and has consequently received increasing attention in recent years.

Early studies explored the applicability of CNNs to RSISR. Liebel and Körner [22] pioneered the use of CNNs for enhancing the spatial resolution of remote sensing data, laying the foundation for subsequent deep learning-based approaches. Lei et al. [23] proposed LGCNet, which exploits deep hierarchical features to improve reconstruction accuracy. Huang et al. [24] introduced Remote Sensing Deep Residual Learning (RS-DRL), a residual learning-based framework that effectively enhances fine-grained details in RSIs. Haut et al. [25] developed the deep compendium model (DCM), which combines residual blocks and skip connections to jointly capture local structures and global contextual information. To further improve multiscale representation, Wang et al. [26] presented Attention-Based Multilevel Feature Fusion Network (AMFFN), an attention-driven multilevel feature fusion network that achieves a favorable tradeoff between reconstruction accuracy and computational efficiency. Wang et al. [27], [28] presented EGSRN, an edge-guided SR framework that leverages explicit edge cues and dedicated fusion mechanisms to improve the preservation of structural boundaries in RSIs.

In recent years, researchers have increasingly introduced transformer architectures into RSISR tasks to compensate for the limitations of conventional convolutional models in modeling global information and long-range spatial dependencies. TransENet [29] employed a transformer architecture to better capture complex spatial correlations in RSIs. Kang et al. [30] proposed ESTNet, an efficient Swin Transformer-based network that integrates channel attention to enhance structural detail preservation while maintaining computational efficiency. Xiao et al. [31], [32], [33] introduced TTST, a Top-k token selective transformer that significantly reduces computational cost while achieving superior reconstruction performance. In addition, diffusion-based models, such as EDiffSR [33], have been explored to mitigate oversmoothing effects commonly observed in CNN-based RSISR methods, demonstrating improved detail recovery and visual quality. In the broader image SR community, Wu et al. [34] proposed SeeSR, which trains a degradation-aware prompt extractor to generate semantic prompts for guiding pretrained text-to-image diffusion models in real-world image SR. Beyond visual feature modeling in RSISR, Chen et al. [35] proposed SeG-SR, which introduces semantic guidance by extracting CLIP-based semantic features from LR images and generating semantic guidance maps to modulate different SR units. However, the existing semantic-guided SR methods still mainly rely on semantic prompts or implicit visual semantic embeddings, while explicit textual semantic priors and direct image–text feature interaction for progressive RSISR remain insufficiently explored.

Unlike the existing RSISR methods that predominantly rely on visual information and focus on improving pixel-level reconstruction quality, the proposed CTF-SRNet explicitly incorporates textual semantic priors to guide the SR process. By introducing cross-modal text and feature modulation, the network leverages high-level semantic information to complement low-level visual cues, enabling more semantically consistent and structurally reliable reconstruction. Moreover, the proposed image–text feature interaction mechanism employs cross-attention fusion between original image features and text-modulated features, effectively alleviating semantic drift while preserving structural fidelity. This design allows CTF-SRNet to better handle complex remote sensing scenes with diverse objects and heterogeneous textures, leading to improved reconstruction quality and enhanced semantic consistency.

III. Proposed Method

In this section, we first describe the overall architecture of the proposed CTF-SRNet. Subsequently, we provide detailed explanations of the image encoder, ITFIB, and FMB modules. Finally, an in-depth discussion related to CTF-SRNet is presented.

A. Overview of CTF-SRNet

In this article, we propose a CTF-SRNet for RSISR. As illustrated in Fig. 1, the CTF-SRNet is designed to incorporate textual semantic priors into the SR process through cross-modal feature interaction and progressive optimization. The proposed framework consists of three major components: a multiscale channel-modulated image encoder, an ITFIB, and a POM for hierarchical reconstruction. By jointly exploiting visual information and textual guidance, the CTF-SRNet aims to enhance structural fidelity and texture reconstruction in complex remote sensing scenes.

The overall workflow of the proposed method is described as follows.

First, LR RSI I_{LR} is fed into the image encoder to extract the initial visual feature representation

$$\boldsymbol{F}_I = Image\ Encoder\,(I_{LR})\,. \tag{1}$$

The image encoder adopts a multiscale channel modulation strategy to capture structural and textural information at different spatial scales, providing a robust visual foundation for subsequent cross-modal interaction.

Before being fed into the text encoder, the textual semantic message is constructed offline by a frozen Generative Pretrained Transformer/Vision-Language Model-based captioning model (GPT/VLM-based captioning model). Specifically, for each LR image, we use the frozen captioning model to generate a natural-language description that summarizes the main object category, visual attributes, and salient structural information of the remote sensing scene. The generated textual descriptions are neither manually annotated nor derived from category labels. Instead, they are directly generated from the LR images and saved before training and evaluation. To avoid information leakage, only LR images are used for caption generation, while HR/GT images are not involved in the construction of textual semantic priors.

The generated textual semantic message can be formulated as

$$\mathcal{T}_{LR} = \mathcal{G}_{cap}\,(I_{LR}) \tag{2}$$

where $\mathcal{G}_{cap}$ denotes the frozen GPT/VLM-based captioning model used only for offline caption generation.

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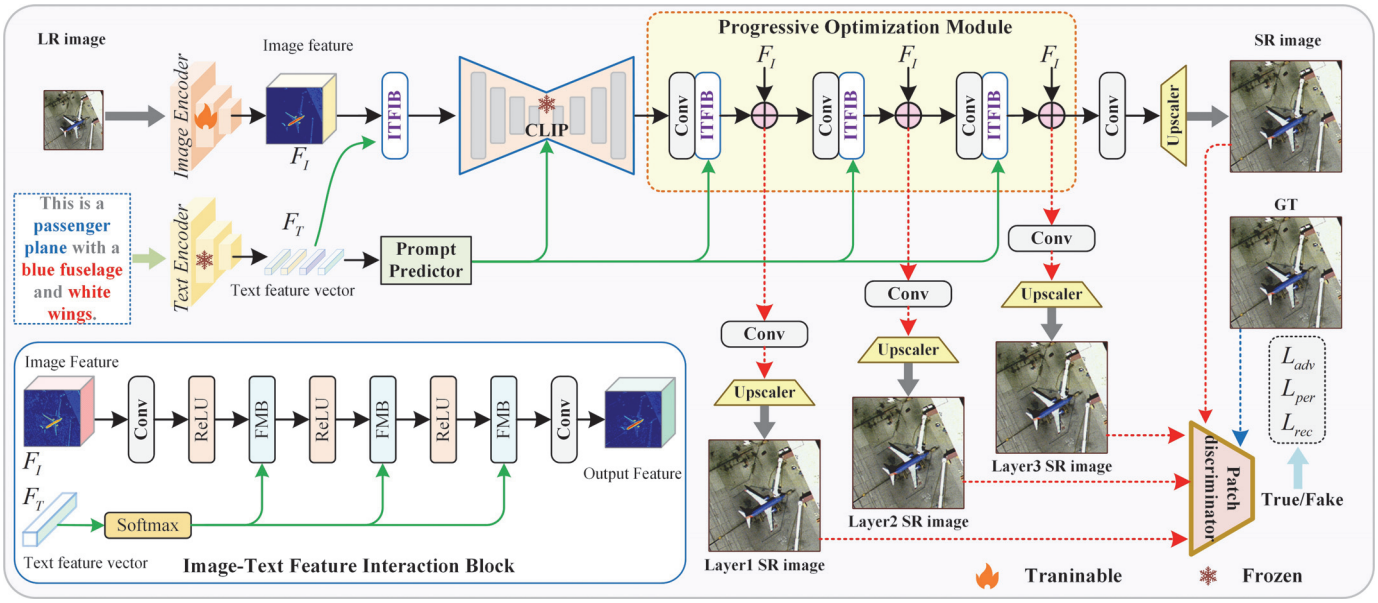


Fig. 1. Overall architecture of the proposed CTF-SRNet. The network consists of a multiscale channel-modulated image encoder, an image–text feature interaction module, and a progressive reconstruction optimization module. By introducing CLIP-based textual guidance, the ITFIB and FMB enable effective fusion of image features and textual information. The POM iteratively refines the SR results under multilevel supervision, leading to improved structural and texture reconstruction quality.

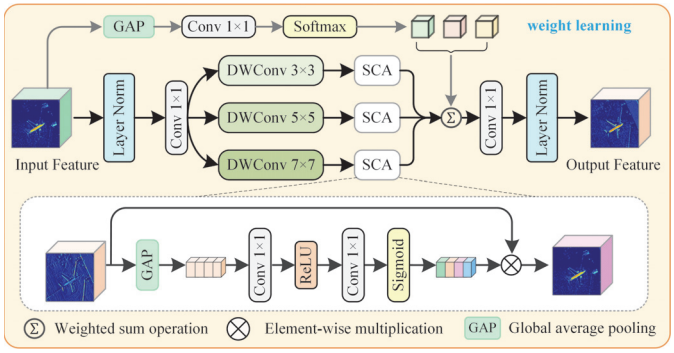


Fig. 2. Architecture of the multiscale channel-modulated image encoder.

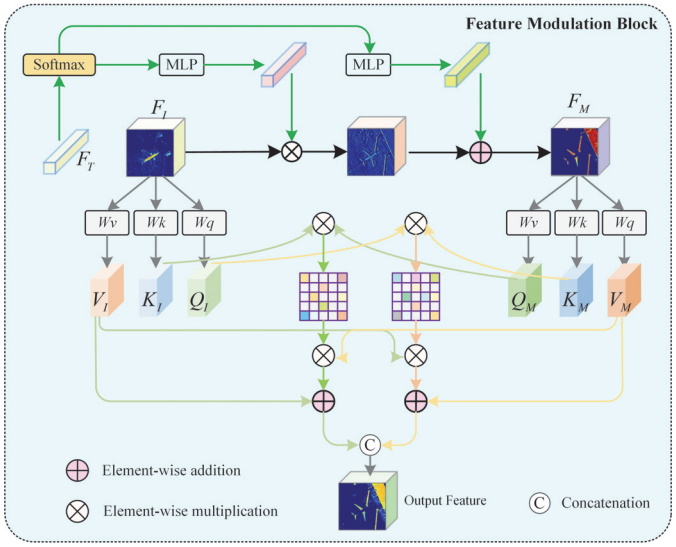


Fig. 3. Illustration of FMB.

Meanwhile, the corresponding pregenerated textual semantic message is encoded by a text encoder to obtain the textual feature vector

$$f_T = Text\ Encoder\left(\mathcal{T}_{LR}\right). \tag{3}$$

It is worth noting that the textual semantic prior is used in both training and testing. However, during the inference stage, the GPT/VLM captioning model is not invoked online. The proposed CTF-SRNet directly loads the pregenerated textual description or its encoded textual feature. Therefore, the offline captioning model does not introduce additional trainable parameters to the proposed SR network.

The extracted image features and textual features are then jointly processed by the ITFIB. In addition, the CLIP-based cross-modal alignment is employed to further integrate textual semantic information into the visual feature space. The CLIP learns a unified image–text embedding space through large-scale vision–language pretraining, enabling effective semantic alignment between visual representations and textual descriptions. In RSISR, image degradation often leads to significant loss of structural and semantic information, making it difficult to accurately recover HR details from visual features alone. By leveraging the scene-level contextual information and rich semantic knowledge encoded in CLIP, the proposed framework establishes effective connections between visual features and semantic priors, providing more reliable semantic guidance for the reconstruction process. Consequently, the reconstructed results exhibit improved structural consistency and semantic plausibility. The initial fused representation is formulated as

$$\boldsymbol{F}_0 = CLIP\left(ITFIB\left(\boldsymbol{F}_I, \boldsymbol{f}_T\right), \boldsymbol{f}_T\right). \tag{4}$$

Subsequently, the fused features are progressively refined through a series of ITFIB modules within the POM. At each stage, the features are first transformed by a convolutional

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layer and then interact with the textual features. To preserve the original visual information, a residual connection with the corresponding image feature F_I is introduced

$$\boldsymbol{F}_i = ITFIB\left(Conv\left(\boldsymbol{F}_{i-1}\right), \boldsymbol{f}_T\right) + \boldsymbol{F}_I, \quad i = 1, 2, 3. \tag{5}$$

At each intermediate stage, the refined features are projected to the image space via convolution and upsampling operations to generate multilevel SR outputs, enabling hierarchical supervision during training

$$\text{Layer}_i \text{ SR image} = \text{Upscaler}\left(\text{Conv}\left(\boldsymbol{F}_i\right)\right), \quad i = 1, 2, 3. \tag{6}$$

Finally, the output of the last stage is further processed to produce the final super-resolved image

$$I_{SR} = \text{Upscaler}\left(\text{Conv}\left(\boldsymbol{F}_3\right)\right). \tag{7}$$

Through progressive feature refinement and multilevel supervision, CTF-SRNet gradually enhances structural details and texture fidelity, while effectively leveraging textual semantic information to guide the SR reconstruction.

B. Image Encoder

As shown in 2, the image encoder is designed as a multiscale channel-modulated feature extraction module to capture structural and textural information at different spatial scales. Given the input feature F_{input}, the encoder first applies layer normalization followed by a convolutional transformation to obtain an initial unified feature representation

$$\boldsymbol{U}_0 = \text{Conv}\left(\text{LayerNorm}\left(\boldsymbol{F}_{input}\right)\right). \tag{8}$$

To model multiscale spatial context, the normalized features U_0 are processed by multiple depthwise convolution branches with different kernel sizes. Specifically, three depthwise convolutions with kernel sizes 3×3, 5×5, and 7×7 are employed to capture local, medium-range, and large-scale spatial structures, respectively. Each branch is further enhanced by a spatial-channel attention (SCA) mechanism to emphasize informative responses

$$\boldsymbol{U}_i = \text{SCA}\left(\text{DWConv}_i\left(\boldsymbol{U}_0\right)\right), \quad i \in \{3, 5, 7\}. \tag{9}$$

Meanwhile, to adaptively balance the contributions of multiscale features, a channelwise weight learning mechanism is introduced. The input feature F_{input} is first aggregated by global average pooling (GAP), followed by a 1×1 convolution and a softmax operation to generate normalized channel weights

$$\boldsymbol{U}_{\text{channel}_k} = \text{Softmax}\left(\text{Conv}\left(\text{GAP}\left(\boldsymbol{F}_{input}\right)\right)\right), \quad k = 1, 2, 3. \tag{10}$$

The learned channel weights are then used to modulate the corresponding multiscale feature branches. Specifically, each scale-specific feature U_{i_k} is combined with its associated channel weight through elementwise addition

$$\boldsymbol{U}_n = \boldsymbol{U}_{\text{channel}_k} + \boldsymbol{U}_{i_k}, \quad k = 1, 2, 3,\ i_k \in \{3, 5, 7\}. \tag{11}$$

Finally, the fused multiscale features are compressed by a convolutional layer and normalized to produce the output feature representation of the image encoder

$$F_{output} = \text{LayerNorm}\left(\text{Conv}\left(U_n\right)\right). \tag{12}$$

Through multiscale depthwise convolution and channel-adaptive modulation, the proposed image encoder effectively captures diverse structural patterns and texture distributions in RSIs, providing robust visual features for subsequent cross-modal interaction and progressive SR reconstruction.

C. Image–Text Feature Interaction Block

The image–text feature interaction block (ITFIB) is designed to integrate textual semantic information into visual feature representations in a progressive and controllable manner. Given the image feature F_I extracted by the image encoder and the corresponding text feature f_T, ITFIB performs feature interaction by modulating image features with text-driven guidance while preserving the original visual structure.

First, the text feature f_T is normalized by a softmax operation to generate a semantic guidance vector, which serves as a global modulation signal

$$\boldsymbol{w}_T = \text{Softmax}\left(\boldsymbol{f}_T\right). \tag{13}$$

The image feature F_I is then processed by a convolutional layer and fed into a stack of FMBs. Each FMB injects the textual guidance into the image feature space through feature modulation, enabling cross-modal interaction between visual and textual information

$$\boldsymbol{f}_I^{(l)} = \text{FMB}\left(ReLU\left(\boldsymbol{f}_I^{(l-1)}\right), w_T\right), \quad l = 1, 2, \ldots, L \tag{14}$$

where $F_I^{(0)} = \text{Conv}(F_I)$ and L denotes the number of stacked FMBs. Through this progressive modulation process, textual semantics are gradually incorporated into the visual features, enhancing semantic awareness without disrupting structural representations.

After feature modulation, the output features are further refined through a convolutional transformation to produce the final output of ITFIB

$$F_{\text{out}} = \text{Conv}\left(\boldsymbol{F}_I^{(L)}\right). \tag{15}$$

By repeatedly applying feature modulation guided by textual information, the ITFIB enables effective image–text interaction at the feature level. This design allows semantic cues to complement visual features in a controlled manner, providing enriched representations for subsequent progressive SR reconstruction.

D. Feature Modulation Block

As shown in Fig. 3, The FMB is designed to perform fine-grained interaction between visual features and textual semantics, serving as the core unit of the ITFIB. Given the image feature F_I and the corresponding text feature f_T, the FMB injects textual guidance into the visual feature space through feature modulation and cross-attention-based fusion, enabling effective cross-modal representation learning.

First, the text feature f_T is normalized by a softmax operation and mapped by a multilayer perceptron (MLP) to generate a modulation vector

$$M_T = \text{MLP}\left(\text{Softmax}\left(\boldsymbol{f}_T\right)\right). \tag{16}$$

The modulation vector is then applied to the image feature via elementwise multiplication to produce a text-modulated visual feature

$$\hat{\boldsymbol{F}}_I = \boldsymbol{F}_I \odot M_T. \tag{17}$$

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To further enhance the modulated representation, an additional MLP is employed to generate a complementary bias term, which is combined with the modulated feature through elementwise addition

$$\boldsymbol{F}_M = \hat{F}_I \oplus \mathrm{MLP}(M_T) \tag{18}$$

where F_M denotes the modulated image feature enriched by textual semantics. To jointly exploit the original and modulated features, the FMB adopts a cross-attention fusion mechanism. Specifically, the original image feature F_I and the modulated feature F_M are, respectively, projected into query, key, and value spaces

$$Q_I = W_q F_I, \quad K_I = W_k F_I, \quad V_I = W_v F_I \tag{19}$$

$$Q_M = W_q F_M, \quad K_M = W_k F_M, \quad V_M = W_v F_M. \tag{20}$$

Based on these projections, the bidirectional cross-attention is computed to capture mutual dependencies between the two feature representations

$$A_{I\to M} = \mathrm{Softmax}\left(\frac{Q_I K_M^\top}{\sqrt{d}}\right) \tag{21}$$

$$A_{M\to I} = \mathrm{Softmax}\left(\frac{Q_M K_I^\top}{\sqrt{d}}\right). \tag{22}$$

The attended features are then obtained as

$$\tilde{F}_I = A_{I\to M} V_M, \quad \tilde{F}_M = A_{M\to I} V_I. \tag{23}$$

Finally, the attended features are combined with their corresponding inputs through residual addition and concatenated to form the output of the FMB

$$\boldsymbol{F}_{\mathrm{out}} = \mathrm{Concat}\left(\tilde{F}_I \oplus F_I,\ \tilde{F}_M \oplus F_M\right). \tag{24}$$

Through text-driven modulation and cross-attention-based feature fusion, the proposed FMB enables effective interaction between visual structures and textual semantics, producing enriched feature representations for subsequent progressive SR reconstruction.

E. Multilevel Loss Functions

To effectively supervise the progressive reconstruction process, we introduce a multilevel loss strategy that jointly constrains intermediate and final SR outputs. By applying hierarchical supervision at different reconstruction stages, the proposed loss design facilitates stable optimization and enables progressive refinement of structural details and textures.

Specifically, the POM produces multiple intermediate super-resolved images at different stages, denoted as $\{I_{SR}^{(i)}\}_{i=1}^{3}$, as well as the final output I_{SR}. Each intermediate output corresponds to a distinct feature refinement stage and is independently supervised during training.

For all intermediate and final outputs, we adopt the ℓ_1 loss as the reconstruction loss to measure the pixelwise discrepancy between the reconstructed image and the ground-truth HR image I_{GT}. Compared with ℓ_2 loss, the ℓ_1 loss is more robust to outliers and is effective in preserving structural information. The reconstruction loss is formulated as

$$\mathcal{L}_{rec} = \sum_{i=1}^{3} \left\| I_{SR}^{(i)} - I_{GT} \right\|_1 + \|I_{SR} - I_{GT}\|_1. \tag{25}$$

To further improve perceptual quality and texture fidelity, a perceptual loss is introduced based on a pretrained Visual Geometry Group (VGG) network. Instead of relying solely on pixel-level supervision, the perceptual loss enforces consistency between reconstructed images and the ground truth in a high-level feature space. It is defined as

$$\mathcal{L}_{per} = \sum_{i=1}^{3} \left\| \phi\left(I_{SR}^{(i)}\right) - \phi\left(I_{GT}\right) \right\|_1 + \|\phi(I_{SR}) - \phi(I_{GT})\|_1 \tag{26}$$

where $\phi(\cdot)$ denotes the feature extractor of the pretrained VGG network.

In addition, to encourage the generation of visually realistic textures, an adversarial loss is incorporated based on a PatchGAN discriminator. The discriminator aims to distinguish between real HR images and generated super-resolved images, while the generator is optimized to fool the discriminator. The adversarial loss is expressed as

$$\mathcal{L}_{adv} = \sum_{i=1}^{3} \mathbb{E}\left[\log\left(1 - D\left(I_{SR}^{(i)}\right)\right)\right] + \mathbb{E}\left[\log\left(1 - D\left(I_{SR}\right)\right)\right]. \tag{27}$$

The overall training objective is defined as a weighted combination of the above loss terms

$$\mathcal{L}_{total} = \lambda_{rec}\mathcal{L}_{rec} + \lambda_{per}\mathcal{L}_{per} + \lambda_{adv}\mathcal{L}_{adv} \tag{28}$$

where λ_{rec}, λ_{per}, and λ_{adv} are the weighting factors that balance the contributions of different loss components.

Through multilevel supervision and joint optimization, the proposed loss strategy effectively guides the network to progressively refine super-resolved images, leading to improved structural fidelity and perceptual quality.

IV. Experimental Results and Analyses

In this section, we comprehensively evaluate the performance of the proposed CTF-SRNet (cross-modal text and feature modulation network) on the task of RSISR. The proposed method is compared with several state-of-the-art RSISR approaches on multiple remote sensing datasets. We first present quantitative and qualitative experimental results to demonstrate the effectiveness of CTF-SRNet in reconstructing HR RSIs. Subsequently, the extensive analyses are conducted to evaluate the robustness and generalization ability of the proposed framework across different datasets and scaling factors. Finally, a series of ablation studies are performed to investigate the contributions of key components, including the image encoder, the image–text feature interaction module, and the multilevel supervision strategy.

A. Datasets

In this work, we employ four representative remote sensing datasets to comprehensively evaluate the performance and generalization capability of the proposed CTF-SRNet. These datasets cover diverse scene types, spatial resolutions, and imaging conditions, providing a solid benchmark for RSISR.

1) AID: The AID [44] is a large-scale benchmark widely used for aerial scene understanding. It contains over 10 000 red, green, blue (RGB) images with a fixed spatial resolution of 600 × 600 pixels, spanning 30 scene categories, such as airport, farmland, industrial, and residential areas. In our

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experiments, 5000 images are randomly selected to construct the training set. For testing, we further sample ten images from each category in the remaining portion of the dataset, resulting in a total of 300 test images.

2) UC Merced Land Use Dataset: The UC Merced Land Use dataset [48] consists of 2100 aerial RGB images with a spatial resolution of 256 × 256 pixels, covering 21 land-use categories, including residential, forest, agricultural, freeway, and parking lot scenes. This dataset is commonly adopted for evaluating SR performance under relatively small-scale but diverse scene conditions. Following standard practice, the dataset is evenly divided into training and testing subsets.

3) RSSCN7 Dataset: The RSSCN7 dataset [49] contains 2800 RSIs with a spatial resolution of 400 × 400 pixels, covering seven typical scene categories, such as grassland, residential, industrial, and river. The images are collected under varying weather, seasonal, and illumination conditions, making RSSCN7 suitable for evaluating the robustness of SR models. Similar to UC Merced, the dataset is evenly split into training and testing sets, with a portion of the training data reserved for validation.

4) DOTA: The DOTA [50] provides 11 268 large-scale aerial images collected from multiple platforms, including Google Earth, Gaofen-2 satellite, Jilin-1 satellite, and airborne sensors. The image resolutions range from 800 × 800 to over 20 000 × 20 000 pixels, covering complex scenes with diverse object distributions. To assess the generalization ability of the proposed method on multisource and large-scale data, the DOTA is used exclusively for testing and is excluded from the training process. For computational efficiency, the images with resolutions larger than 2000 × 2000 pixels are discarded, and 250 images are randomly selected to form the test set.

For all datasets, the original HR images are treated as ground truth. The corresponding LR images are generated using bicubic downsampling with different scale factors. All images are converted to portable network graphics (PNG) format and paired to construct the HR–LR datasets required for training and evaluation.

B. Evaluation Metrics

To quantitatively evaluate the SR performance, we adopt peak signal-to-noise ratio (PSNR) and structural similarity index measure (SSIM) [51] as the primary evaluation metrics. Following common practice in RSISR, all quantitative results are reported on the luminance (Y) channel in the YCbCr color space. In addition to reconstruction accuracy, we further assess the computational characteristics of different methods by comparing their numbers of parameters and floating-point operations (FLOPs).

C. Training Settings

During training, data augmentation techniques, including random rotations and horizontal flips, are employed to improve model robustness. The proposed network is optimized using the Adam optimizer, where 64 × 64 patches are randomly cropped from LR images as training samples. The model is trained for a total of 1000 K iterations with a batch size of 64. The initial learning rate is set to 1×10^{-3} and is gradually decayed at predefined milestones of [100, 500, 800, 900, and 950 K]. All experiments are implemented in PyTorch and conducted on a single NVIDIA RTX 4090 GPU.

D. Quantitative Results

We quantitatively evaluate the proposed CTF-SRNet against a wide range of the state-of-the-art SR methods on multiple remote sensing datasets, including UC Merced, RSSCN7, AID, DOTA, and DIOR. The compared methods cover classical CNN-based models (e.g., EDSR [15] and RDN [36]), attention-based and transformer-based approaches (e.g., RCAN [16], HAN [37], HAT [19], RGT [41], and MambaIR [42]), as well as recent remote sensing-oriented SR methods (e.g., EGSRN [27] and FMSR [43]). In addition, for a fairer comparison with text-guided SR methods, we further include SeeSR [34], a representative semantics-aware SR approach, as an additional baseline. The quantitative comparisons are conducted under multiple upscaling factors, and performance is evaluated using PSNR, SSIM, and Learned Perceptual Image Patch Similarity (LPIPS) metrics.

1) Results on UC Merced and RSSCN7: The quantitative results on the UC Merced and RSSCN7 datasets with scale factors ×2, ×3, and ×4 are summarized in Table I. As shown, the CTF-SRNet consistently achieves competitive or superior performance across most settings. On the UC Merced dataset, CTF-SRNet attains 35.51 dB/0.9418 SSIM for ×2 and 30.20 dB/0.8520 SSIM for ×3, demonstrating strong reconstruction capability at moderate scales. For the more challenging ×4 setting, CTF-SRNet achieves 27.73 dB in PSNR and 0.7930 in SSIM, outperforming the majority of compared methods and yielding stable improvements over both CNN-based and transformer-based baselines.

On the RSSCN7 dataset, which contains scenes captured under varying weather and illumination conditions, CTF-SRNet also exhibits robust performance. As reported in Table I, our method achieves 27.97-dB PSNR and 0.7233 SSIM for ×4 SR, surpassing recent methods, such as EGSRN and FMSR. These results indicate that the proposed cross-modal text and feature modulation strategy generalizes well across datasets with diverse imaging conditions and scene complexities.

To further investigate category-level performance, we report the per-class PSNR results for the ×4 setting on UC Merced and RSSCN7 in Tables II and III, respectively. On UC Merced, CTF-SRNet achieves the highest average PSNR of 27.77 dB and shows consistent improvements across most scene categories, particularly in structurally complex classes, such as dense residential, industrial, and storage areas. Similarly, on RSSCN7, CTF-SRNet obtains the best average PSNR of 27.97 dB and outperforms all competing methods across all seven scene categories, demonstrating strong robustness to scene diversity and environmental variations.

2) Results on AID, DOTA, and DIOR: To further assess the generalization capability of CTF-SRNet on larger scale and more diverse remote sensing benchmarks, we conduct additional experiments on the AID, DOTA, and DIOR datasets. The quantitative results are reported in Table IV, where PSNR, SSIM, and LPIPS are jointly considered to evaluate both pixel-level accuracy and perceptual quality.

As shown in Table IV, CTF-SRNet achieves the best overall performance across all three datasets. On the AID, our method attains 31.05-dB PSNR, 0.8189 SSIM, and the lowest LPIPS value of 0.2956, indicating superior reconstruction fidelity and perceptual consistency. Compared with recent frequency-domain and lightweight remote sensing methods, CTF-SRNet consistently delivers improved structural detail recovery.

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TABLE I

QUANTITATIVE RESULTS WITH SCALE FACTORS ×2, ×3, AND ×4 ON UC MERCED AND RSSCN7 DATASETS. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Method	Scale	UC Merced	RSSCN7	Scale	UC Merced	RSSCN7	Scale	UC Merced	RSSCN7
		PSNR/SSIM	PSNR/SSIM		PSNR/SSIM	PSNR/SSIM		PSNR/SSIM	PSNR/SSIM
Bicubic	×2	29.07/0.8509	28.84/0.7974	×3	26.47/0.7587	27.93/0.7306	×4	25.33/0.6651	26.23/0.6498
EDSR [15]		33.62/0.9124	31.87/0.8821		29.21/0.8287	28.94/0.7834		27.03/0.7359	27.28/0.6996
RDN [36]		33.78/0.9151	31.95/0.8840		29.32/0.8314	29.02/0.7852		27.14/0.7398	27.36/0.7025
RCAN [16]		33.91/0.9180	32.05/0.8873		29.45/0.8348	29.11/0.7886		27.26/0.7441	27.45/0.7059
HAN [37]		33.97/0.9192	32.09/0.8884		29.52/0.8365	29.17/0.7903		27.32/0.7460	27.50/0.7076
HSENet [38]		33.88/0.9171	32.02/0.8862		29.41/0.8339	29.08/0.7878		27.21/0.7429	27.42/0.7042
NLSN [39]		34.01/0.9203	32.12/0.8897		29.57/0.8382	29.21/0.7920		27.36/0.7488	27.56/0.7095
HAUNet [40]		34.08/0.9216	32.18/0.8910		29.64/0.8397	29.27/0.7935		27.41/0.7509	27.61/0.7112
TransENet [29]		33.95/0.9189	32.07/0.8879		29.48/0.8356	29.13/0.7894		27.28/0.7451	27.47/0.7068
HAT [19]		34.12/0.9231	32.22/0.8934		29.70/0.8426	29.32/0.7960		27.49/0.7563	27.70/0.7149
RGT [41]		34.15/0.9292	32.55/0.8992		29.95/0.8456	29.43/0.7982		27.56/0.7610	27.84/0.7184
MambaIR [42]		34.23/0.9310	32.61/0.9013		30.01/0.8473	29.49/0.8004		27.62/0.7645	27.89/0.7201
EGSRN [27]		34.31/0.9332	32.68/0.9017		30.09/0.8498	29.55/0.8027		27.69/0.7679	27.92/0.7220
SeeSR [34]		34.16/0.9214	32.36/0.8896		29.94/0.8431	29.45/0.7968		27.52/0.7615	27.79/0.7162
FMSR [43]		34.42/0.9360	32.79/0.9025		30.18/0.8519	29.58/0.8037		27.67/0.7672	27.94/0.7231
CTF-SRNet (Ours)		35.51/0.9418	32.82/0.9028		30.20/0.8520	29.61/0.8040		27.77/0.7930	27.97/0.7233

TABLE II

UC MERCED TEST DATASET AVERAGE PSNR (DB) FOR EACH CLASS (×4). THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Class	EDSR [15]	RDN [36]	HAT [19]	RGT [41]	MambaIR [42]	SeeSR [34]	FMSR [43]	**CTF-SRNet (Ours)**
1	26.56	26.82	27.03	26.75	26.59	26.82	26.94	27.08
2	28.85	28.85	28.76	28.80	28.84	28.84	28.91	29.16
3	31.65	31.62	31.64	31.67	31.65	31.59	31.70	32.74
4	34.74	34.78	34.74	34.76	34.74	34.72	34.77	34.85
5	25.86	25.90	25.82	25.79	25.85	25.88	25.93	26.17
6	26.03	25.95	25.99	26.00	26.03	26.00	26.03	26.45
7	26.31	26.38	26.37	26.31	26.81	26.35	26.39	26.50
8	28.63	28.66	28.61	28.68	28.61	28.60	28.65	28.70
9	32.38	32.37	32.35	32.37	32.38	32.42	32.65	32.79
10	23.49	23.45	23.46	23.39	23.37	23.55	23.67	23.86
11	25.12	25.20	25.09	25.02	25.26	25.18	25.25	25.26
12	26.68	26.71	26.71	26.73	26.84	26.76	26.77	26.89
13	27.08	27.11	27.12	27.14	27.13	27.22	27.31	27.37
14	28.43	28.47	28.42	28.42	28.47	28.50	28.54	28.68
15	29.72	29.76	29.79	29.81	29.76	29.75	29.78	30.06
16	29.13	29.14	29.15	29.14	29.14	29.20	29.29	29.43
17	30.94	30.95	30.96	30.97	31.05	30.90	30.98	31.17
18	28.56	28.55	28.60	28.59	28.57	28.56	28.61	28.97
19	30.08	30.07	30.07	30.09	30.12	30.07	30.06	30.35
20	28.83	28.87	28.86	28.88	28.83	28.82	28.95	28.91
21	28.38	28.38	28.43	28.45	28.46	28.42	28.44	28.47
AVG	27.03	27.14	27.49	27.56	27.62	27.52	27.73	27.77

TABLE III

RSSCN7 TEST DATASET AVERAGE PSNR (DB) FOR EACH CLASS (×4). THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Class	EDSR [15]	RDN [36]	HAT [19]	RGT [41]	MambaIR [42]	SeeSR [34]	FMSR [43]	**CTF-SRNet (Ours)**
1	32.86	32.85	32.86	32.83	32.85	32.78	32.89	32.94
2	32.23	32.23	32.24	32.21	32.23	32.14	32.26	32.31
3	24.82	24.85	24.81	24.80	24.81	24.78	24.86	24.90
4	29.85	29.87	29.84	29.86	29.85	29.80	29.88	29.92
5	27.10	27.11	27.10	27.10	27.09	27.04	27.14	27.18
6	24.04	24.06	24.02	24.03	24.02	23.99	24.07	24.11
7	24.08	24.10	24.07	24.08	24.08	24.00	24.12	24.16
AVG	27.28	27.36	27.70	27.84	27.89	27.79	27.94	27.97

On the DOTA, which consists of large-scale aerial images collected from multiple sensors and platforms, CTF-SRNet achieves 34.18-dB PSNR, 0.8732 SSIM, and 0.2291 LPIPS, outperforming all compared methods by a clear margin. This demonstrates the effectiveness of the proposed cross-modal text-guided feature modulation in handling complex spatial layouts and multiscale object distributions. Similarly, on the DIOR dataset, CTF-SRNet achieves 31.10-dB PSNR, 0.8215 SSIM, and 0.2932 LPIPS, further confirming its robustness and strong generalization ability across datasets with rich semantic diversity.

To provide a more detailed analysis, we further present categorywise results on the AID under the ×4 setting in Table V. It can be observed that CTF-SRNet consistently achieves the best or second-best performance across almost all scene categories. In particular, the notable improvements are observed in categories, such as airport, industrial, residential, storage tanks, and viaduct, which typically contain complex structures and fine-grained textures. These results highlight the advantage of incorporating textual semantic guidance into the SR process for enhancing structure-aware and semantically consistent detail reconstruction.

Overall, the comprehensive quantitative results across five remote sensing datasets demonstrate that CTF-SRNet achieves stable and consistent performance improvements under different scaling factors, evaluation metrics, and scene types. By effectively integrating cross-modal textual information with image feature modulation and multilevel supervision, the proposed method exhibits strong reconstruction accuracy, perceptual quality, and generalization capability for RSISR tasks.

3) Qualitative Results: Figs. 4 and 5 present the qualitative comparisons of the ×4 SR results on the UC Merced, RSSCN7, AID, and DOTA datasets. From the visual results, it can be observed that the proposed CTF-SRNet consistently reconstructs sharper edges, clearer structural boundaries, and more faithful textures than the existing state-of-the-art methods.

On the UC Merced dataset, the representative scenes, such as *Runway* and *Storage Tanks*, are shown in Fig. 4. For the runway example, the conventional CNN- and transformer-based methods, such as EDSR, RCAN, and HAT, tend to produce blurred strip patterns and suffer from edge smearing,

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TABLE IV

QUANTITATIVE COMPARISON ON AID, DOTA, AND DIOR DATASETS. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Methods	Venue	AID [44]			DOTA [45]			DIOR [46]			Average		
		PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
Bicubic	–	28.86	0.7382	0.4803	31.16	0.7947	0.4043	28.57	0.7432	0.4678	29.53	0.7587	0.4508
EDSR [15]	CVPR'2017	30.65	0.8086	0.3068	33.64	0.8648	0.2616	30.63	0.8116	0.3020	31.64	0.8283	0.2901
RDN [36]	CVPR'2018	30.74	0.8112	0.3157	33.60	0.8670	0.2421	30.78	0.8147	0.3093	31.71	0.8310	0.2890
RCAN [16]	ECCV'2018	30.82	0.8121	0.3112	33.86	0.8680	0.2384	30.85	0.8159	0.3048	31.84	0.8320	0.2848
HAN [37]	ECCV'2020	30.80	0.8122	0.3079	33.84	0.8682	0.2363	30.84	0.8163	0.3026	31.83	0.8322	0.2823
HSENet [38]	TGRS'2022	30.72	0.8108	0.3124	33.85	0.8667	0.2412	30.77	0.8143	0.3070	31.78	0.8306	0.2869
NLSN [39]	CVPR'2021	30.81	0.8126	0.3044	33.86	0.8682	0.2349	30.82	0.8156	0.3004	31.83	0.8321	0.2799
HAUNet [40]	TGRS'2023	30.88	0.8132	0.3111	33.94	0.8687	0.2403	30.87	0.8160	0.3074	31.90	0.8326	0.2863
TransENet [29]	TGRS'2022	30.80	0.8109	0.3136	33.75	0.8675	0.2418	30.85	0.8148	0.3075	31.80	0.8311	0.2881
HAT [19]	CVPR'2023	30.81	0.8124	0.3078	33.99	0.8684	0.2392	30.87	0.8161	0.3062	31.89	0.8323	0.2844
GRL [47]	CVPR'2023	30.86	0.8127	0.3085	33.86	0.8710	0.2372	30.90	0.8177	0.3047	31.87	0.8338	0.2835
RGT [41]	ICLR'2024	30.91	0.8159	0.3023	33.92	0.8709	0.2337	30.91	0.8182	0.2992	31.91	0.8350	0.2784
MambaIR [42]	ECCV'2024	30.85	0.8130	0.3086	33.86	0.8691	0.2377	30.89	0.8167	0.3060	31.87	0.8329	0.2845
EGSRN [27]	TGRS'2025	30.92	0.8158	0.3035	33.98	0.8705	0.2356	30.95	0.8180	0.3008	31.95	0.8345	0.2800
SeeSR [34]	CVPR'2024	30.89	0.8147	0.3041	33.93	0.8698	0.2351	30.92	0.8174	0.3012	31.91	0.8340	0.2801
FMSR [43]	TMM'2024	30.93	0.8156	0.3001	34.04	0.8710	0.2325	30.97	0.8187	0.2983	31.98	0.8353	0.2770
CTF-SRNet (Ours)	–	31.05	0.8189	0.2956	34.18	0.8732	0.2291	31.10	0.8215	0.2932	32.11	0.8379	0.2726

TABLE V

CATEGORYWISE RESULTS ON THE AID. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Categories	Bicubic		EDSR [15]		HSENet [38]		NLSN [39]		HAT [19]		MambaIR [42]		RGT [41]		FMSR [43]		CTF-SRNet (Ours)	
	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑
Airport	27.83	0.7554	29.93	0.8282	30.08	0.8303	30.16	0.8322	30.15	0.8319	30.21	0.8327	30.29	0.8352	30.26	0.8346	30.38	0.8374
Bare Land	35.60	0.8564	36.94	0.8837	36.79	0.8841	37.00	0.8845	36.88	0.8841	36.95	0.8843	37.00	0.8852	37.02	0.8847	37.14	0.8873
Baseball Field	31.00	0.8305	33.05	0.8765	33.15	0.8774	33.24	0.8787	33.25	0.8789	33.32	0.8792	33.35	0.8805	33.43	0.8803	33.40	0.8830
Beach	32.90	0.8446	34.18	0.8727	34.14	0.8746	34.31	0.8749	34.34	0.8756	34.36	0.8754	34.41	0.8766	34.38	0.8759	34.52	0.8787
Bridge	30.22	0.8283	32.93	0.8800	33.06	0.8809	33.12	0.8818	33.04	0.8809	33.14	0.8821	33.22	0.8832	33.28	0.8836	33.41	0.8860
Center	26.51	0.6944	28.77	0.7921	28.83	0.7937	28.95	0.7971	28.92	0.7956	28.99	0.7966	29.09	0.8015	29.09	0.8010	29.22	0.8038
Church	24.29	0.6333	26.30	0.7469	26.47	0.7507	26.51	0.7528	26.56	0.7532	26.61	0.7543	26.66	0.7580	26.64	0.7567	26.78	0.7601
Commercial	27.33	0.7174	29.01	0.7940	29.21	0.7989	29.21	0.7996	29.21	0.8007	29.29	0.8012	29.30	0.8048	29.34	0.8039	29.46	0.8070
D-Residential	22.93	0.5671	24.38	0.6839	24.60	0.6912	24.60	0.6936	24.67	0.6936	24.74	0.6965	24.78	0.7019	24.78	0.7019	24.92	0.7045
Desert	39.26	0.9100	40.20	0.9268	39.57	0.9271	40.27	0.9278	40.37	0.9278	40.09	0.9270	40.15	0.9283	40.29	0.9275	40.46	0.9301
Farmland	33.10	0.8226	35.00	0.8683	35.02	0.8692	35.10	0.8699	35.03	0.8691	35.09	0.8696	35.14	0.8715	35.18	0.8711	35.30	0.8738
Forest	28.79	0.6605	29.85	0.7315	30.00	0.7363	29.98	0.7369	30.01	0.7363	30.00	0.7370	30.01	0.7400	30.04	0.7391	30.16	0.7395
Industrial	26.77	0.6952	28.88	0.7931	28.98	0.7956	29.04	0.7982	29.04	0.7980	29.09	0.7988	29.20	0.8034	29.17	0.8027	29.32	0.8060
Meadow	33.86	0.7483	34.63	0.7804	34.55	0.7804	34.69	0.7821	34.70	0.7815	34.64	0.7811	34.69	0.7831	34.74	0.7827	34.86	0.7855
M-Residential	26.36	0.6335	28.34	0.7365	28.45	0.7390	28.49	0.7418	28.46	0.7408	28.59	0.7435	28.66	0.7479	28.65	0.7472	28.80	0.7502
Mountain	29.51	0.7349	30.63	0.7885	30.72	0.7907	30.74	0.7916	30.78	0.7923	30.76	0.7918	30.71	0.7941	30.78	0.7928	30.92	0.7954
Park	29.06	0.7530	30.54	0.8130	30.71	0.8167	30.71	0.8177	30.71	0.8189	30.76	0.8185	30.81	0.8211	30.80	0.8204	30.94	0.8234
Parking	24.24	0.7060	27.25	0.8317	27.32	0.8341	27.57	0.8408	27.56	0.8405	27.65	0.8414	27.79	0.8472	28.02	0.8524	28.18	0.8550
Playground	32.64	0.8450	35.37	0.8943	35.46	0.8952	35.58	0.8967	35.49	0.8959	35.54	0.8961	35.65	0.8978	35.72	0.8978	35.84	0.8975
Pond	30.70	0.8167	32.11	0.8542	32.17	0.8549	32.22	0.8559	32.18	0.8555	32.24	0.8559	32.24	0.8567	32.23	0.8560	32.36	0.8585
Port	26.67	0.7986	28.50	0.8596	28.71	0.8623	28.71	0.8631	28.81	0.8638	28.83	0.8644	28.90	0.8667	28.92	0.8668	29.05	0.8692
Railway Station	26.78	0.6793	28.72	0.7738	28.84	0.7762	28.89	0.7783	28.88	0.7780	28.99	0.7802	29.03	0.7833	29.04	0.7830	29.18	0.7856
Resort	26.79	0.7029	28.52	0.7799	28.64	0.7825	28.68	0.7845	28.71	0.7849	28.76	0.7857	28.80	0.7889	28.82	0.7886	28.96	0.7913
River	30.37	0.7402	31.55	0.7891	31.61	0.7904	31.64	0.7914	31.63	0.7909	31.64	0.7912	31.67	0.7927	31.66	0.7918	31.79	0.7944
School	27.41	0.7237	29.36	0.8044	29.51	0.8074	29.55	0.8097	29.54	0.8104	29.62	0.8113	29.71	0.8150	29.70	0.8143	29.84	0.8173
S-Residential	26.66	0.6006	27.71	0.6728	27.84	0.6754	27.84	0.6767	27.88	0.6759	27.88	0.6768	27.91	0.6796	27.91	0.6789	28.05	0.6818
Square	28.55	0.7391	30.84	0.8200	30.94	0.8223	31.04	0.8244	31.00	0.8251	31.05	0.8247	31.16	0.8284	31.15	0.8276	31.12	0.8304
Stadium	27.16	0.7547	29.63	0.8387	29.68	0.8391	29.79	0.8422	29.77	0.8422	29.83	0.8421	29.94	0.8457	29.97	0.8460	30.10	0.8486
Storage Tanks	25.65	0.6793	27.44	0.7664	27.58	0.7688	27.61	0.7709	27.60	0.7698	27.64	0.7709	27.71	0.7744	27.71	0.7739	27.85	0.7766
Viaduct	26.97	0.6755	28.99	0.7757	29.08	0.7772	29.17	0.7813	29.11	0.7794	29.19	0.7810	29.26	0.7851	29.26	0.7840	29.40	0.7868
Average	28.86	0.7382	30.65	0.8086	30.72	0.8108	30.81	0.8126	30.81	0.8124	30.85	0.8130	30.91	0.8159	30.93	0.8156	31.05	0.8189

especially around high-contrast markings. Although SeeSR introduces semantic guidance, it may generate excessive local textures and unstable structural details due to the domain gap between natural image Real-ISR and RSISR. In contrast, CTF-SRNet successfully recovers clear runway stripes with well-defined boundaries and uniform spacing. Similarly, in the storage tank scene, our method better preserves the circular geometry and sharp outlines of tank structures, while other methods exhibit noticeable ringing artifacts, oversmoothed boundaries, or loss of fine details.

For the RSSCN7 dataset, Fig. 4 illustrates the challenging scenes containing dense parking areas and complex man-made structures. Most competing approaches struggle to reconstruct thin line structures and small objects, leading to texture distortion and oversmoothing. The SeeSR can introduce additional visual details, but some of them are not well aligned with the original remote sensing structures, resulting in less stable reconstruction around repetitive parking lines and vehicle contours. Benefiting from text-guided feature modulation and progressive refinement, CTF-SRNet produces visually cleaner

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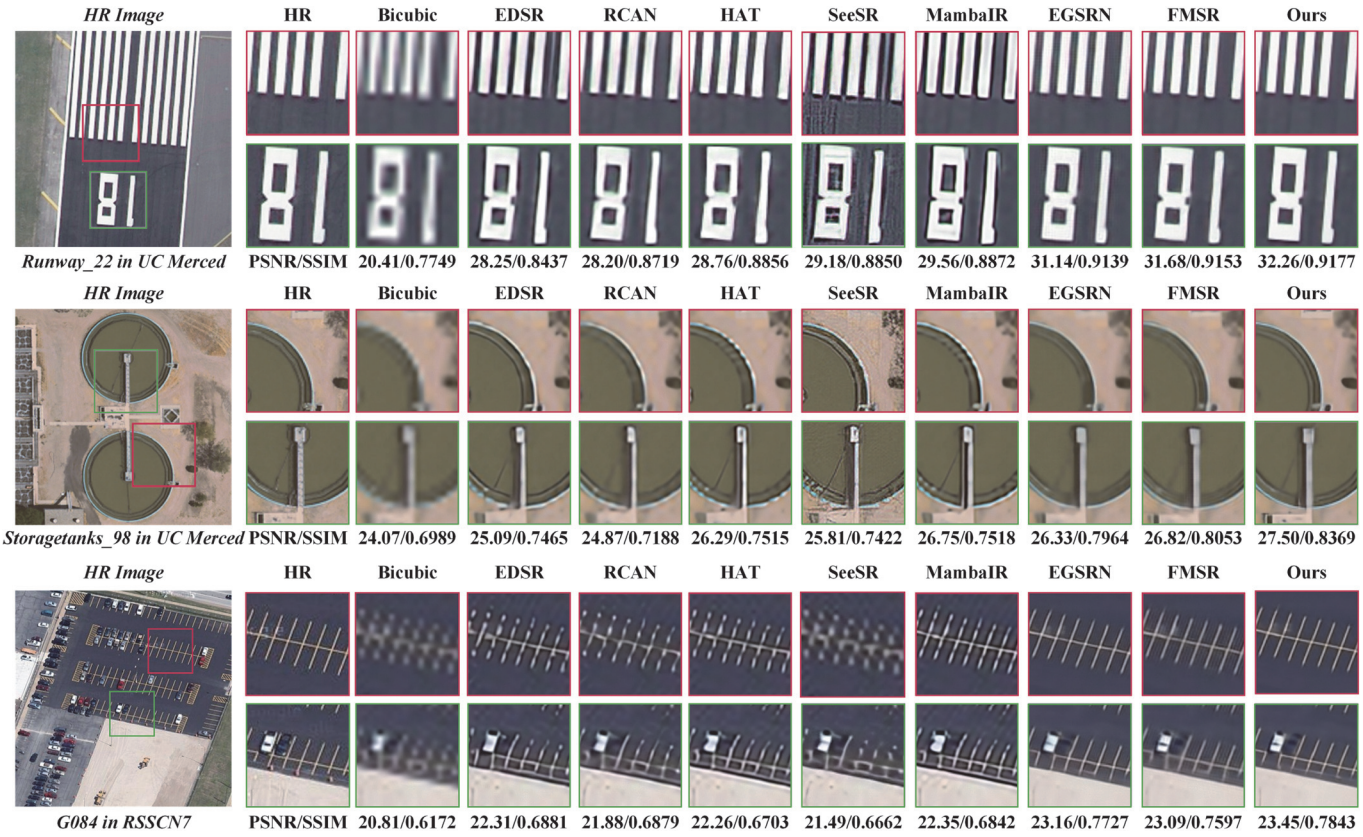


Fig. 4. Result comparison of our method with other SR methods on the UC Merced and RSSCN7 test datasets (4×).

results with enhanced structural continuity, more accurate line patterns, and reduced artifacts.

The qualitative results on the AID are shown in Fig. 5, including *stadium* and *parking* scenes. These scenes are characterized by repetitive patterns and fine-grained textures. While advanced transformer-based models, such as HAT and MambaIR, improve overall sharpness, they still fail to fully restore subtle line structures and local contrast. Compared with the general semantics-aware SR method SeeSR, CTF-SRNet better adapts textual semantic guidance to remote sensing scenes and avoids introducing hallucinated textures. As a result, our method reconstructs clearer lane markings, sharper field boundaries, and more coherent textures, yielding results that are visually closer to the ground truth.

Furthermore, to evaluate the generalization capability of the proposed method, we present qualitative comparisons on the DOTA dataset in Fig. 5. Since DOTA images are collected from diverse sensors and were not used during training, they pose significant challenges due to real-world degradations and complex scene layouts. Despite these difficulties, CTF-SRNet effectively restores fine edges and structural details, such as road markings and field patterns, while suppressing noise and blur. This demonstrates the strong robustness and generalization ability of our method on unseen remote sensing data.

To further evaluate the robustness and cross-domain generalization capability of the proposed method, we additionally conduct qualitative experiments on the Houston hyperspectral dataset. Different from conventional RGB remote sensing datasets, the hyperspectral images contain richer spectral information and usually exhibit more complex imaging characteristics, which poses additional challenges for SR models. As shown in Fig. 6, CTF-SRNet can still recover clearer spatial structures from LR Houston images, such as building boundaries, road layouts, sports fields, and vegetation regions. The enlarged regions show that our method is able to enhance local structural details and improve the visual clarity of real hyperspectral scenes, indicating that the proposed cross-modal feature modulation strategy has promising generalization ability beyond standard RGB RSIs.

However, we also observe that some fine-grained details in the Houston hyperspectral scenes are not fully recovered. For example, small objects, subtle boundary transitions, and highly mixed texture regions may still suffer from blurring or incomplete reconstruction. This is mainly because hyperspectral images have more complex spectral–spatial characteristics than RGB images, while the current framework primarily focuses on spatial SR with textual semantic guidance. These observations suggest that although CTF-SRNet shows effectiveness on real hyperspectral scenarios, further improvements are still needed. In future work, we will explore spectral–spatial joint modeling and hyperspectral-specific semantic guidance to better preserve both spatial structures and spectral consistency.

Overall, these qualitative comparisons confirm that CTF-SRNet achieves superior visual quality across diverse remote sensing scenarios. Compared with both image-only SR methods and the representative semantics-aware SeeSR, the proposed method more effectively enhances high-frequency

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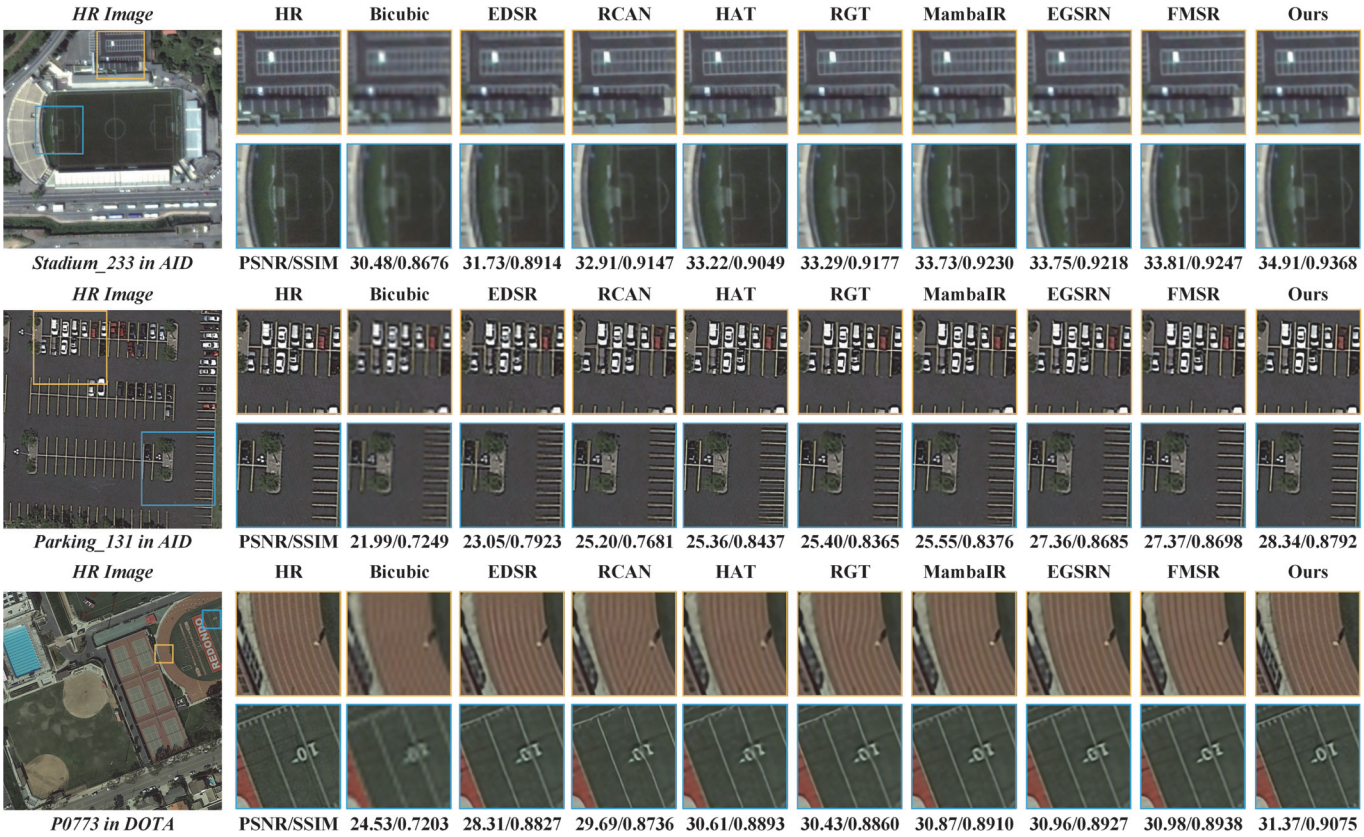


Fig. 5. Result comparison of our method with other SR methods on the AID and DOTA test dataset (4×).

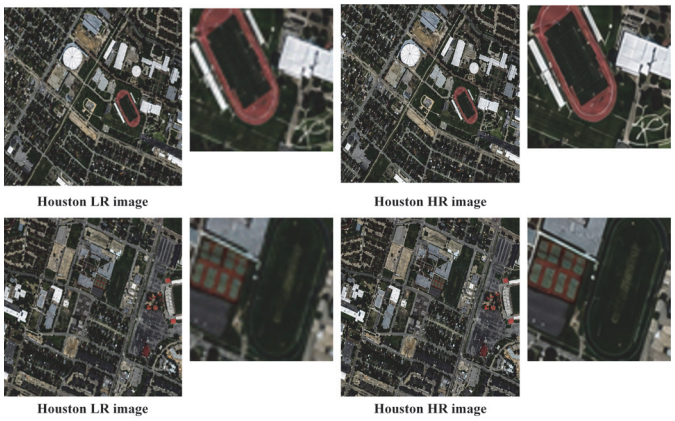


Fig. 6. Comparison between the LR image and the CTF-SRNet SR reconstruction results on the Houston hyperspectral dataset.

details, preserves structural integrity, and maintains semantic consistency in complex remote sensing scenes.

E. Ablation Study

To comprehensively analyze the contribution of each key component in the proposed CTF-SRNet, we conduct detailed ablation experiments on the UC Merced dataset under the ×4 SR setting. The quantitative results are summarized in Table VI, while the qualitative comparisons and attention analyses are illustrated using local attribution maps (LAMs) [52] in Fig. 7.

TABLE VI
ABLATION STUDY ON THE UC MERCED TEST DATASET FOR ×4 SR

Variant	Params	M-Adds	PSNR	SSIM	Time
Baseline	0.52M	18.40G	26.94	0.7638	18.30ms
Without Image Encoder	0.39M	13.10G	26.31	0.7485	15.40ms
Without ITFIB	0.61M	21.75G	27.21	0.7724	20.10ms
Without FMB	0.68M	24.90G	27.46	0.7812	22.35ms
Without Multi-level Loss	0.81M	28.60G	27.61	0.7875	24.80ms
CTF-SRNet (Ours)	0.81M	28.60G	**27.73**	**0.7930**	24.80ms

The baseline model corresponds to a simplified network without the image encoder, ITFIB, FMB, and multilevel loss supervision, serving as a reference configuration.

1) Effect of the Image Encoder: The image encoder is designed to extract multiscale and channel-adaptive representations from LR inputs. As shown in Table VI, removing the image encoder leads to a noticeable performance degradation, with PSNR dropping from 27.73 to 26.31 dB and SSIM decreasing from 0.7930 to 0.7485, despite reduced computational cost.

From the LAM visualization in Fig. 7, the model without the image encoder exhibits weak and scattered activation responses, with a low difference intensity (DI) value of 3.0250. In contrast, the baseline already shows more concentrated responses (DI = 1.5530), indicating that the image encoder effectively enhances low-level structural sensitivity and provides a stronger foundation for subsequent feature modulation.

In addition to evaluating the overall contribution of the image encoder, we further conduct an internal ablation study

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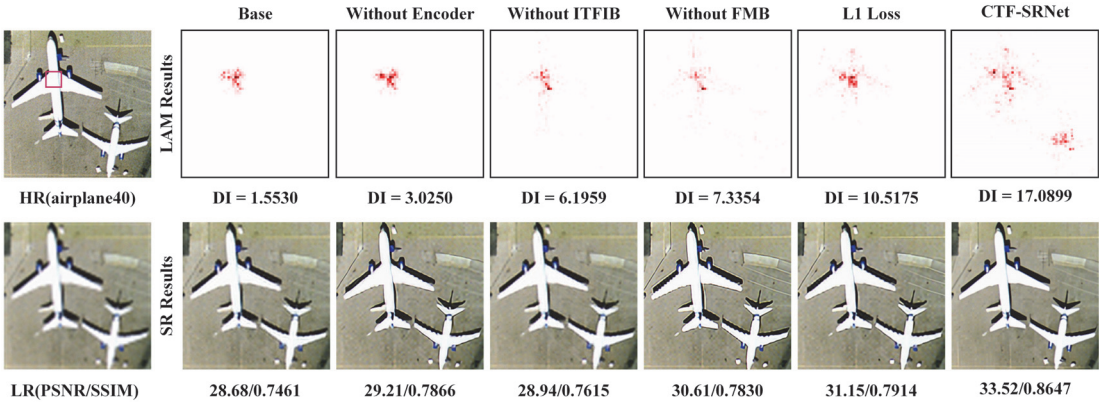


Fig. 7. Comparative analysis of reconstruction quality and LAM-based attribution across different model variants on UC Merced (4× SR).

TABLE VII

ABLATION STUDY OF THE PROPOSED IMAGE ENCODER ON THE UC MERCED DATASET FOR ×4 SR. HERE, ✓ DENOTES THAT THE CORRESPONDING COMPONENT IS USED, WHILE "–" DENOTES THAT IT IS REMOVED. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Setting	Image Encoder Components			Performance	
	Multi-scale Branches	SCA	Weight Learning	PSNR↑	SSIM↑
Conv 3 × 3	–	✓	✓	23.43	0.7558
w/o SCA	✓	–	✓	23.21	0.7485
w/o Weight Learning	✓	✓	–	22.97	0.7396
Full Encoder	✓	✓	✓	**23.86**	**0.7683**

TABLE VIII

ABLATION STUDY OF DIFFERENT MULTILEVEL SUPERVISION STRATEGIES ON THE UC MERCED DATASET FOR ×4 SR. HERE, "–" DENOTES THAT THE CORRESPONDING INTERMEDIATE OUTPUT IS NOT SUPERVISED, WHILE ✓ DENOTES THAT THE OUTPUT IS SUPERVISED. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Setting	Level	Intermediate Supervision			Final	Performance	
		Layer1	Layer2	Layer3		PSNR↑	SSIM↑
S1	1-level	–	–	✓	✓	28.16	0.8315
S2	2-level	–	✓	✓	✓	28.75	0.8491
S3	3-level	✓	✓	✓	✓	**29.11**	**0.8571**

TABLE IX

ABLATION STUDY OF DIFFERENT SUPERVISION-LEVEL WEIGHT COMBINATIONS ON THE UC MERCED DATASET FOR ×4 SR. THE WEIGHTS REPRESENT THE RELATIVE CONTRIBUTIONS OF DIFFERENT INTERMEDIATE RECONSTRUCTION LAYERS IN THE MULTILEVEL SUPERVISION STRATEGY. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Setting	Supervision Weights			Performance	
	Layer1	Layer2	Layer3	PSNR↑	SSIM↑
S1	0.25	0.25	0.25	26.55	0.6410
S2	0.50	0.50	0.50	27.27	0.6915
S3	1.00	1.00	1.00	27.32	0.6963
S4	0.25	0.50	0.75	27.63	0.7133
S5	0.50	0.75	1.00	**27.91**	**0.7240**

TABLE X

ABLATION STUDY OF DIFFERENT LOSS-WEIGHT SETTINGS ON THE UC MERCED DATASET FOR ×4 SR. HERE, "–" DENOTES THAT THE CORRESPONDING LOSS TERM IS NOT USED, WHILE ✓ DENOTES THAT THE LOSS TERM IS USED. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Setting	Loss Components			Loss Weights			Performance	
	L_{rec}	L_{per}	L_{adv}	λ_{rec}	λ_{per}	λ_{adv}	PSNR↑	SSIM↑
S1	✓	–	–	1.0	0	0	22.41	0.7143
S2	✓	✓	–	1.0	0.05	0	22.32	0.7114
S3	✓	–	✓	1.0	0	0.005	22.16	0.7043
S4	–	✓	✓	0	0.05	0.005	19.90	0.5794
S5	✓	✓	✓	1.0	0.01	0.005	21.73	0.6865
S6	✓	✓	✓	1.0	0.10	0.005	22.02	0.6980
S7	✓	✓	✓	1.0	0.05	0.001	22.11	0.6992
S8	✓	✓	✓	1.0	0.05	0.010	22.24	0.7078
S9	✓	✓	✓	1.0	0.05	0.005	**23.86**	**0.7683**

to analyze the effectiveness of its key components. As reported in Table VII, replacing the proposed multiscale branches with a single Conv 3 × 3 operation leads to a decrease in performance, achieving 23.43-dB PSNR and 0.7558 SSIM. This indicates that the multiscale convolutional branches are beneficial for capturing remote sensing structures with different spatial extents, such as narrow roads, building edges, and large homogeneous regions.

When the SCA module is removed, the performance drops to 23.21-dB PSNR and 0.7485 SSIM. This demonstrates that SCA plays an important role in emphasizing informative spatial and channel responses while suppressing less relevant features. Without this attention mechanism, the encoder becomes less effective in extracting discriminative structural and textural representations.

Furthermore, removing the weight learning branch results in the lowest performance among the three internal variants, with 22.97-dB PSNR and 0.7396 SSIM. This suggests that the GAP–Conv–Softmax pathway is crucial for adaptively balancing the contributions of different scale branches. Directly fusing multiscale features without learned channelwise weights may introduce redundant or less relevant responses, thereby weakening the representation capability of the encoder.

By integrating all components, the full image encoder achieves the best performance, with 23.86-dB PSNR and 0.7683 SSIM. These results verify that the multiscale branches, SCA module, and adaptive weight learning mechanism are complementary to each other, jointly improving the encoder's ability to extract robust and structure-aware visual features for RSISR.

2) Impact of the ITFIB: The ITFIB is introduced to inject text-guided semantic cues into the image feature stream. As reported in Table VI, removing ITFIB results in a PSNR of

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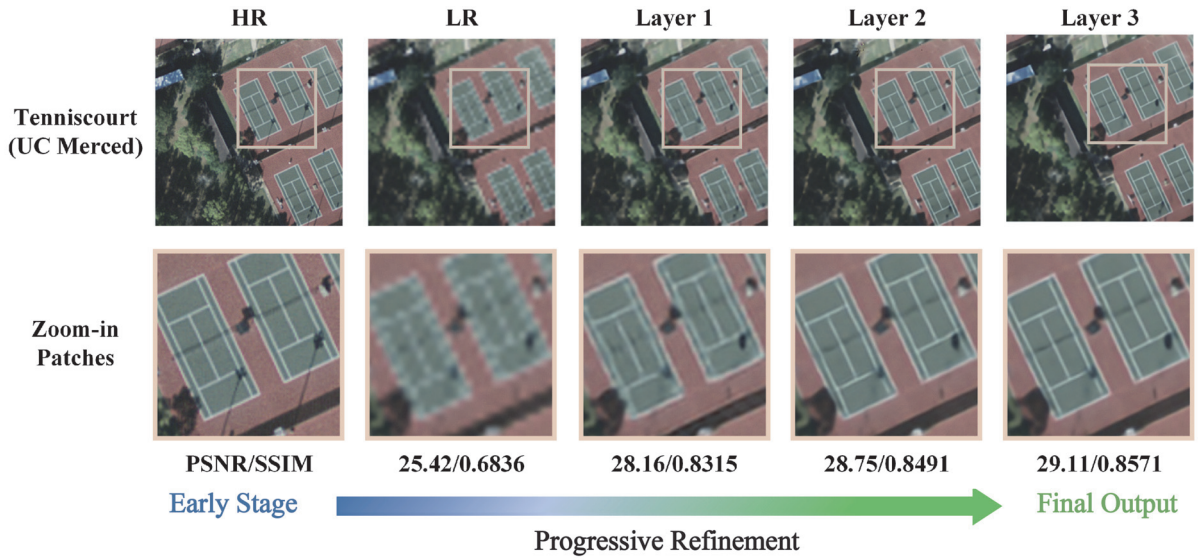


Fig. 8. Visualization of progressive reconstruction results from different layers of CTF-SRNet on the UC Merced dataset for $\times 4$ SR.

27.21 dB and an SSIM of 0.7724, which are significantly lower than those of the full model.

The LAM results further support this observation. As shown in Fig. 7, the DI value increases from 3.0250 (without Encoder) to 6.1959 when ITFIB is present, indicating that ITFIB effectively guides the model's attention toward semantically meaningful regions, such as object contours and distinctive structures. Without ITFIB, the attention distribution becomes more diffused and less aligned with target structures.

3) Contribution of the FMB: The FMB is designed to adaptively refine feature responses via cross-modal modulation. As shown in Table VI, removing FMB leads to a performance drop to 27.46-dB PSNR and 0.7812 SSIM.

The LAM visualizations reveal a corresponding decrease in structural sensitivity. The DI value for the model without FMB is 7.3354, which is noticeably lower than that of the full model. This indicates that FMB plays a critical role in reinforcing discriminative features and suppressing irrelevant background responses, especially around fine-grained edges and object boundaries.

4) Effectiveness of the Multilevel Loss Strategy: The proposed multilevel loss strategy supervises intermediate SR outputs at different stages, enabling progressive optimization. When this strategy is removed, the PSNR and SSIM decrease to 27.61 dB and 0.7875, respectively, as shown in Table VI.

The LAM results show a DI value of 10.5175 under single-level (L1) supervision, which is higher than those of models without ITFIB or FMB but still inferior to the full model. This demonstrates that multilevel supervision effectively enhances structural consistency and improves attention concentration on high-frequency regions.

To further investigate the influence of different supervision levels on model performance, we conduct an ablation study of one-, two-, and three-level multilevel supervision strategies, with the quantitative results reported in Table VIII. The one-level supervision strategy only supervises the Layer3 intermediate reconstruction output together with the final output, achieving 28.16-dB PSNR and 0.8315 SSIM. By additionally introducing supervision on Layer2, the two-level strategy improves the performance to 28.75-dB PSNR and 0.8491 SSIM. When Layer1, Layer2, and Layer3 intermediate reconstruction outputs are all supervised, the proposed three-level strategy achieves the best performance, reaching 29.11-dB PSNR and 0.8571 SSIM. Furthermore, the intermediate reconstruction results at different layers are visualized in Fig. 8, where the reconstruction quality can be observed to progressively improve from Layer1 to Layer3. Early stages mainly recover coarse structural information, while later stages further enhance boundary details and texture information. These results demonstrate that multilevel supervision effectively facilitates progressive feature refinement. Considering that introducing additional supervision levels requires extra reconstruction branches and increases computational complexity, the three-level supervision strategy provides a better tradeoff between reconstruction performance and computational cost and is therefore adopted as the default setting in this work.

Furthermore, to investigate the influence of relative weighting among different intermediate supervision levels, we conduct an additional ablation study with different supervision weight combinations. The quantitative results are reported in Table IX, and the corresponding visual comparisons are shown in Fig. 9. As shown in Table IX, different weight assignments among Layer1, Layer2, and Layer3 lead to different reconstruction performances. For equal weighting strategies, the performance gradually improves with increasing supervision strength. Specifically, S1 (0.25, 0.25, 0.25), S2 (0.50, 0.50, 0.50), and S3 (1.00, 1.00, 1.00) achieve 26.55 dB/0.6410, 27.27 dB/0.6915, and 27.32 dB/0.6963 in terms of PSNR/SSIM, respectively. However, simply increasing the supervision strength of all intermediate layers does not achieve the optimal performance. By assigning larger weights to deeper reconstruction layers, S4 (0.25, 0.50, 0.75) and S5 (0.50, 0.75, 1.00) further improve the reconstruction quality. Among all settings, S5 achieves the best performance, reaching 27.91-dB PSNR and 0.7240 SSIM. As illustrated in Fig. 9, S5 reconstructs clearer boundaries and more complete structural details compared with other weighting strategies.

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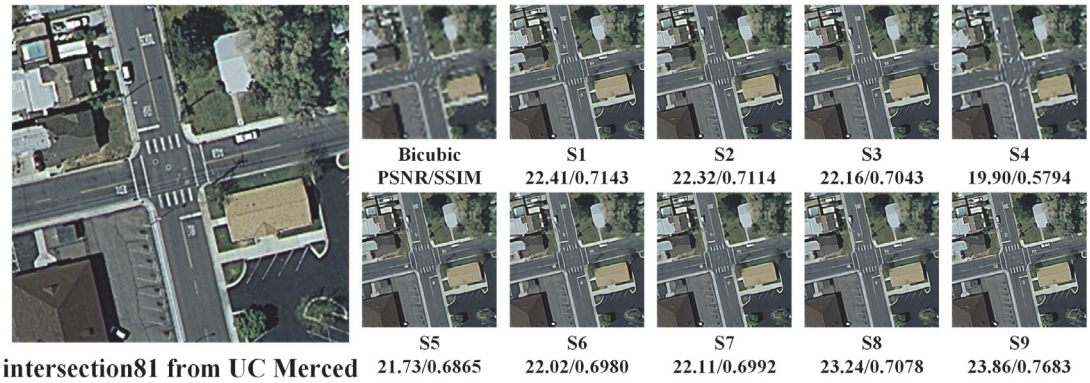


Fig. 9. Qualitative ablation results of different loss-weight settings on the UC Merced dataset for ×4 SR.

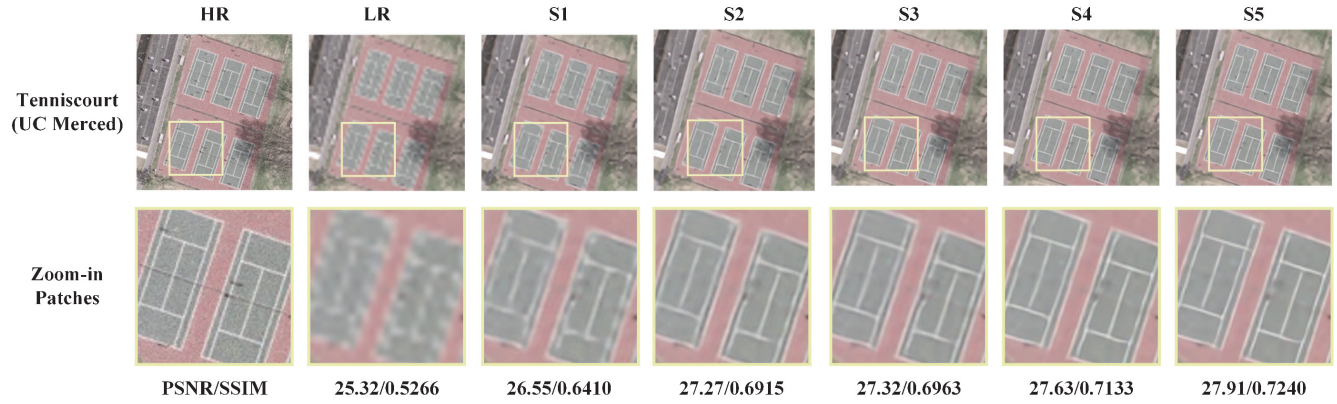


Fig. 10. Visual comparison of different supervision-level weight combinations on the UC Merced dataset for ×4 SR.

These results indicate that deeper reconstruction layers contain more high-frequency information and benefit from stronger supervision, while appropriate supervision on early layers facilitates stable progressive feature refinement. Therefore, the progressive supervision weighting strategy provides a better balance between intermediate feature optimization and final reconstruction quality.

Furthermore, we conduct an additional loss-weight ablation study to investigate the influence of different combinations of reconstruction loss, perceptual loss, and adversarial loss. The quantitative results are reported in Table X, and the corresponding qualitative comparison is shown in Fig. 9. As shown in Table X, the setting without reconstruction loss, i.e., S4, obtains the lowest PSNR and SSIM values of 19.90 dB and 0.5794, respectively. This indicates that the reconstruction loss is essential for maintaining pixel-level fidelity and stable structural recovery in RSISR.

When only the reconstruction loss is used, S1 achieves 22.41-dB PSNR and 0.7143 SSIM, providing a relatively strong baseline. However, the visual results still contain blurred road markings and weakened fine structures, as shown in Fig. 10. By introducing perceptual loss or adversarial loss, the network can further enhance local textures and high-frequency details, but inappropriate weighting may lead to unstable reconstruction or structural distortion. For example, S5–S8 show that different weights of perceptual and adversarial losses influence the balance between reconstruction fidelity and perceptual enhancement.

Among all settings, S9, with $\lambda_{rec} = 1.0$, $\lambda_{per} = 0.05$, and $\lambda_{adv} = 0.005$, achieves the best performance, reaching 23.86-dB PSNR and 0.7683 SSIM. As illustrated in Fig. 10, S9 reconstructs clearer lane markings, sharper building boundaries, and more coherent road structures compared with other settings. These results demonstrate that an appropriate combination of reconstruction, perceptual, and adversarial losses can effectively balance pixel-level accuracy and perceptual quality, thereby improving the structural fidelity and visual realism of the super-resolved RSIs.

5) Overall Analysis: By integrating all components, the CTF-SRNet achieves the best overall performance with 27.73-dB PSNR and 0.7930 SSIM, while maintaining a lightweight architecture with only 0.81M parameters. As illustrated in Fig. 7, the full model produces the most concentrated attribution maps with the highest DI value of 17.0899, indicating strong sensitivity to object structures and fine details.

Both quantitative results and LAM-based qualitative analyses consistently demonstrate that each module contributes synergistically to the final reconstruction quality, validating the rationality and effectiveness of the proposed CTF-SRNet.

V. Conclusion

In this article, we propose CTF-SRNet, a novel cross-modal SR framework for RSISR. By explicitly introducing textual semantic priors into the SR process, CTF-SRNet is designed to bridge the gap between high-level scene understanding and

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low-level detail reconstruction, thereby enabling more accurate recovery of structural and semantic details under complex remote sensing scenarios.

The proposed network jointly leverages an image encoder and a text encoder to extract complementary visual and semantic representations, which are interactively fused through the ITFIB. This design allows text-guided semantic cues to dynamically modulate image features, enhancing the model's ability to preserve object structures and scene-specific characteristics. Furthermore, the FMB adaptively refines intermediate features via cross-modal guidance, while a multilevel supervision strategy facilitates progressive optimization and stabilizes training across different reconstruction stages.

Extensive experiments conducted on multiple remote sensing benchmarks, including UC Merced, RSSCN7, AID, and DOTA, demonstrate that CTF-SRNet consistently outperforms the state-of-the-art SR methods in terms of both quantitative metrics and visual quality. In particular, the proposed method shows clear advantages in recovering fine-grained textures, sharp edges, and semantically meaningful structures. The ablation studies and LAM-based analyses further verify the effectiveness and complementary contributions of each key component within the network.

Despite these encouraging results, the current work primarily focuses on RGB remote sensing imagery with predefined textual descriptions. Future work will explore more flexible and automatically generated textual representations, extend the proposed framework to multispectral and hyperspectral data, and investigate blind SR settings. In addition, we plan to further optimize the computational efficiency of CTF-SRNet to support real-time and large-scale remote sensing applications.

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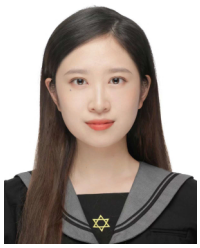
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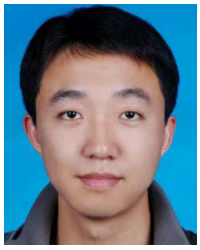
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